

Manned Missions The Indian Chapter

ISRO is awaiting clearance for an ambitious manned space flight with two crew members orbiting the earth by 2015

Along with planetary exploration, ISRO is planning its most ambitious mission: a manned space flight. After years of discussion about the pros and cons, ISRO now says it's ready with a plan. By 2015, it wants to put a two-member crew in a module that would orbit

the earth for a week at a height of 400 km and would splash down into the sea. The cost: Rs 12,000 crore. Its apex decision-making body, the Space Commission, has cleared it and now it awaits government clearance. Rakesh Sharma, India's first man in space, is excited at the prospect. "Humans will soon have to colonise space. It's going to be a great journey and India should be an intimate part of it. That's why we went to Antarctica," he says.

Yet there are plenty of sceptics who not only question the cost but also the need for India to indulge into something like this. Aerospace expert Roddam Narasimha, a Space Commission member, points out: "The difference between India and China is that here we doubt ourselves, the Chinese just go ahead and do it." That has seen China race ahead of India both in planetary mission and manned space flights. Its first manned mission

went up in 2003 and recently its taikonaut undertook the first space walk by a Chinese.

ISRO is not under the illusion that doing a manned mission is going to be easy. As a launcher, the organisation is planning to modify its Geo-synchronous Launch Vehicle, GSLV, which can carry a four-tonne payload to an orbit of 36,000 km. With its proposed crew module weighing around three tonne, the GSLV would be able to inject it into orbit comfortably. But before that ISRO would have to master the technology of building such a module, including the creation of a "shirt-sleeve" environment for the crew that would allow them to live inside it with some comfort. Life support systems would have to be developed, including waste management and disposal.

Re-entry technology would have to be mastered so that when the module returns to earth, it can land safely. The module should also have an alternate emergency escape system for the crew in case there is a major engine failure. Redundant power back-up systems would have to be built. For astronauts, special training facilities are being set up in Bangalore to overcome the tremendous physical and psychological stress that space riders experience. Top ISRO scientists feel the effort is worth it. India would not be left behind in the race for space.



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project director, said, "The lift-off was just the baby being born. In the next few days, we have to nurture it so that it can begin to mature and achieve its objectives."

If all goes well, November 15 is the scheduled date when Chandrayaan would reach its final orbital path of 100 km above the moon. It would then orbit the moon every two hours. Among

Chandrayaan's first tasks would be to eject a 35-kg impactor with Indian tricolours painted on it and then send it hurtling towards the moon. The impactor would crash the lunar surface at high speed, sending out a plume of dust and leaving behind the remains of the first Indian presence on an alien surface. It was former President Dr A.P.J. Abdul Kalam who advised the team to